Calibrating the Decision, Not the Model: Per-Class Thresholds and Argmax Scaling for Training-Free Open-Vocabulary Segmentation of Remote Sensing Imagery

Mid-Project Review

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Presentation Outline

1. Introduction

Open-vocabulary semantic segmentation of aerial imagery. The input is an image *and a list of class names as plain words*; the output is a class for every pixel. **Training-free**: no weights are trained, and the class list can change at inference without retraining.

Our baseline: SegEarth-OV3 (Li *et al.*, 2025) applies SAM 3 to remote sensing. Per class prompt SAM 3 emits a presence score, a dense semantic map and instance masks; SegEarth-OV3 fuses them and thresholds the result. Reproduced on our hardware at 47.38 against a published 47.4.

The contribution, in one sentence

A single confidence threshold is the wrong *shape* for a multi-class pipeline. Fitting **one threshold per class**, and **one scale per class before the argmax**, on the same supervision budget the baseline already spends tuning its single τ — **with no weights trained** — recovers a real gain on every dataset we tested.

2. Motivation — the baseline discards land cover

On LoveDA validation, at SegEarth-OV3's own published $\tau = 0.5$:

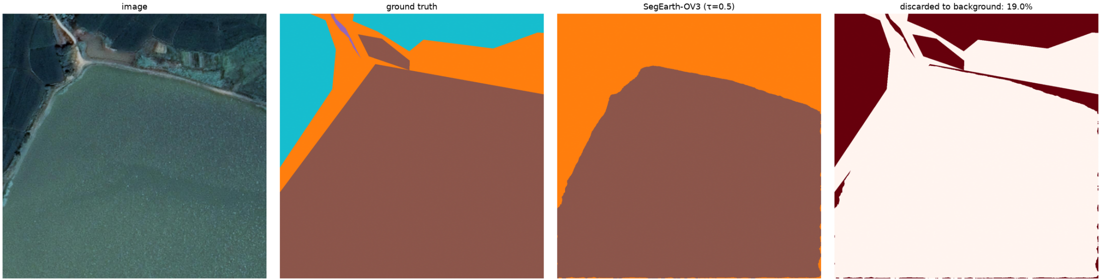
- **29.68%** of all real-class pixels are assigned to `background` — **323 million pixels**.
- Discard outnumbers real-class confusion **3:1**. **The dominant error is silence, not error.**
- `water`: precision **89.5%**, recall **54.7%**. Right nine times in ten when it fires; finds half of what is there.

Every global fix costs more than it returns

lower τ to 0.1	-5.54 mIoU, 1.73 wrong pixels per 1 right
remove presence gating	-11.97 mIoU
do nothing	29.68% of land cover discarded

So the correction has to be selective and per-class, not global. That is the motivation for everything that follows.

2. Motivation — what the discard looks like



A LoveDA tile: **image** / **ground truth** / **SegEarth-OV3 at** $\tau = 0.5$ / **pixels discarded to** **`background`**. Whole water regions are dropped — not misclassified, but declared unknown, because their confidence falls below a threshold tuned for the dataset as a whole rather than for `water`.

2. Applications

Where an open vocabulary is the point

- **Disaster response.** Flood extent from UAV imagery within hours. No labelled data for the affected district, no time to train.
- **Renewable-energy infrastructure.** `solar panel` is a class *no* land-cover benchmark contains — supplied here as two words.
- **Informal settlement mapping.** `tin roof`, `unpaved road`, `water tank`: a vocabulary that changes with every survey.

Measured on Indian UAV imagery

25 tiles, 5 openly licensed scenes, 4–5 cm — a region absent from every benchmark. Run unchanged under two class lists typed at inference.

- The segmentation is stable under renaming: a flooded building partitions the same way under both vocabularies — 78.6% `water` vs 79.1% `flood water`, 18.2% `concrete roof` vs 18.3% `building`.
- And a class with no word is lost: 100.0% of a solar-farm tile goes to one class until `solar panel` is supplied, which recovers 4.6%.

No ground truth exists for this imagery — these are pixel shares, not accuracy.

Cost of deployment

Ours: **∼200 labelled tiles, minutes of CPU, no gradient steps.** Fine-tuning instead:

3. Problem statement — formulation

Let $\mathcal{C} = \{1, \ldots, N\}$ be the vocabulary and $s_c(p) \in [0,1]$ the score the pipeline assigns class c at pixel p.

The baseline decision rule (one global τ, tuned per dataset):

$$\hat{y}(p) = \begin{cases} \arg\max_c s_c(p), & \text{if } \max_c s_c(p) \geq \tau \\ b \ (\textsf{catch-all}), & \text{otherwise} \end{cases}$$

What we fit instead, on ∼200 labelled tiles:

$$\hat{y}(p) = \begin{cases} c^\star, & \text{if } s_{c^\star}(p) \geq \boldsymbol{\tau_{c^\star}} \\ b, & \text{otherwise} \end{cases} \qquad c^\star = \arg\max_c \ \boldsymbol{w_c}\, s_c(p)$$

Objective

$(\boldsymbol{w}, \boldsymbol{\tau}) = \arg\max \mathrm{mIoU}_{\mathrm{real}}$ by coordinate ascent. **No network weights are touched.** The threshold reads the *raw* score, so $\boldsymbol{w}$ acts only on the `argmax`.

3. Why per-class τ is the *complete* family

Claim. Given a fixed `argmax`, any strictly increasing per-class recalibration g_c composed with a global τ collapses to a per-class threshold vector:

$$g_c(s) \;\geq\; \tau \quad\Longleftrightarrow\quad s \;\geq\; g_c^{-1}(\tau) \;=\; \tau_c.$$

- **Per-class τ therefore exhausts everything reachable by rescaling scores after the `argmax`.** Nothing in that family can beat it — which is why the first lever is not an arbitrary choice.
- Scaling *before* the `argmax` changes *which* class wins, so it is a **strictly larger family**.

Measured consequence

6.0% of LoveDA's catch-all assignments are cases where the catch-all *won the **argmax*** at $s \geq \tau$. Lowering a threshold cannot change an `argmax`, so that residual is unreachable by lever one — and it is exactly what lever two addresses.

4. Literature survey

No.	Work	Year	Contribution / relevance
1	**CLIP** [1]	2021	Image–text contrastive pretraining; origin of the open-vocabulary paradigm.
2	**MaskCLIP** [2]	2022	First to extract dense predictions from CLIP by removing pooling.
3	**GroupViT** [3]	2022	Learns segment grouping from text supervision alone.
4	**SCLIP** [4]	2023	Correlative self-attention; the codebase SegEarth-OV3 derives from.
5	**SAM** [5]	2023	Promptable class-agnostic segmentation; masks without a fixed label set.
6	**ClearCLIP** [6]	2024	Isolates which CLIP components harm dense prediction.
7	**ProxyCLIP** [7]	2024	Self-supervised features as structural guidance.
8	**SegEarth-OV** [8]	2025	Feature upsampler + global bias subtraction for remote sensing; defines the evaluation protocol.
9	**SAM 3** [9]	2025	Presence, semantic and instance heads in one promptable model.
10	**SegEarth-OV3** [10]	2025	SAM 3 for RS: dual-head fusion + presence gating. **Our baseline, reproduced at 47.38 vs 47.4.**
11	**ConInfer** [11]	2026	GMM over self-supervised patches fused with the VLM prior. Nearest competitor; used here only as a general

5. Research gaps

1. **The decision rule is unexamined.** Every method in this family ends in `argmax` plus one global τ, tuned per dataset. All effort goes into the *scores*; nobody asks whether the *rule* on top of them has the right shape.
2. **The discarded residual is never quantified.** SegEarth-OV3 reports mIoU; it does not report that **29.68%** of real-class pixels on LoveDA are assigned to `background`.
3. **Catch-all classes distort the metric in both directions.** mIoU is an *unweighted* mean, so a catch-all owns $1/N$ of it however meaningless. We measure it inflating a result on one dataset and deflating one on another.
4. **No published bound on what is achievable without labels.** If a label-free rule could set the thresholds, the calibration cost would vanish. Nobody has tested whether one exists.

6. Objectives

1. **Reproduce** SegEarth-OV3 exactly, so every later number rests on a baseline that runs on our own hardware. ✓ 47.38 vs published 47.4
2. **Quantify** the residual it discards, and test whether any global knob reaches it. ✓ 29.68%; all three knobs cost more than they return
3. **Design** a correction that trains no weights and spends only the supervision budget the baseline already spends. ✓ two levers
4. **Verify** it end-to-end with the official evaluator, not by arithmetic on a cache. ✓ every rung within 0.05 mIoU
5. **Bound** what is achievable with no labels at all. ✓ 12 attempts, all bounded, with a stated mechanism
6. **Demonstrate** on imagery and a vocabulary no benchmark covers. ✓ Indian UAV scenes, 4–5 cm

7. Experimental setup

Datasets — every image used

- **LoveDA** val: **1669** tiles, 1024^2, 0.3 m, 7 classes
- **ISPRS Potsdam**: **2016** tiles, 5 cm, 6 classes
- **OpenEarthMap**: 384 of 500 tiles, 0.25–0.5 m, 9 classes
- **OpenAerialMap (India)**: 25 tiles, no ground truth

Baselines

- **SegEarth-OV3** — reproduced, our primary baseline
- τ sweep, presence removal, per-class Otsu, 12 label-free rules
- ConInfer — a generality check on a different

Protocol

- 5-fold cross-validation; calibration and evaluation tiles **always disjoint**
- The baseline is recomputed on the *same* held-out tiles, so both are scored on identical pixels
- Both metrics reported: full mIoU *and* catch-all-excluded

Hardware

- RTX 2000 Ada, 16 GB, 70 W cap
- One LoveDA pass $\approx$ 24 min
- **Fitting the correction: minutes of CPU**

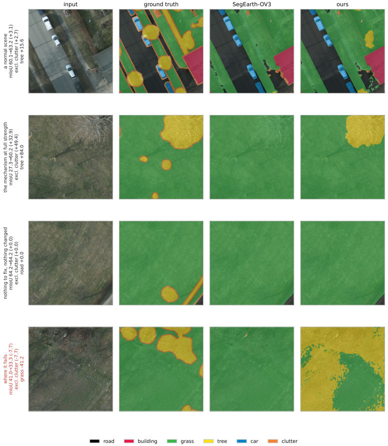
Validation gate: every instrumented run must

7. Results — which dataset, how much

	LoveDA 1669 tiles	**Potsdam** 2016 tiles	**OpenEarthMap** 384 tiles	*ConInfer CLIP*
SegEarth-OV3 (reproduced)	47.38	57.60	44.16	*36.99*
Lever 1: per-class τ	$+1.18 \pm 0.45$	$+0.59$	$+0.30$	$+2.51$
Lever 2: argmax scaling	$+1.16 \pm 0.19$	$+4.86 \pm 0.35$	—	-0.10
Total Δ mIoU	**+2.32**	**+5.67**	+0.30	+2.51
Catch-all-excluded	+1.36	+6.00	**+1.75**	*+1.94*
Folds positive	5/5	5/5	—	5/5

- Verified end-to-end by the official evaluator: Potsdam 57.60 → 58.35 → 63.27 on 1816 held-out tiles, every rung within **0.05** mIoU of its prediction.
- **Potsdam tree: 37.92 → 59.09 IoU** — precision 93.34%, recall 38.63%, exactly the asymmetry the method targets.
- **OpenEarthMap is positive under the recommended metric** (+1.75) even though full mIoU looks flat — its catch-all owns a ninth of the mean and pays for the gain.
- *ConInfer is shown only to answer “is this a SAM 3 quirk?” — it is not the correction transfer to a*

7. Results — the method, tile by tile



Four Potsdam tiles. **Same weights, same forward pass — only the decision rule differs.** A normal urban scene where `tree` improves while already-calibrated classes are left alone; a canopy the baseline misses entirely; a tile the correction barely touches; and a tile where it loses 7.72 mIoU, included deliberately because the five-fold gain $+4.86 \pm 0.35$ is an *average*

7. Honest limitations

- Thresholds do not transfer across a domain shift. Fitted on LoveDA-rural and applied to urban: -1.11 mIoU — *worse than not calibrating at all.* Pooling both domains also loses most of the gain.
- The gain is not uniform. LoveDA rural +2.77, urban +0.10. Our headline is a benchmark-level figure, not a claim about every stratum of it.
- No label-free rule predicts *whether* calibration will pay either, on two datasets. The 200-tile cost buys the answer and the question together.
- The fitted vectors are constant across tiles, so they are right on average and can be wrong on an individual scene.
- OpenEarthMap is 384 of 500 tiles, a public redistribution rather than the official split.

Every negative in this project is reported with an oracle bound, so the reader learns where the remaining headroom is rather than only that we failed to reach it.

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Thank You